

CARMEL POLYTECHNIC COLLEGE, ALAPPUZHA

Government Aided Institution | Approved by AICTE | Affiliated to SBTE, Kerala

AUDIT & EVALUATION REPORT: ATTENDANCE CALCULATION SCHEME FOR CIA / CIE

COMPARATIVE REGULATION AUDIT: REVISION 2021 VS. REVISION 2026

Audit Scope	Attendance Marks Formulation & CIA Aggregation Logic	Audit Date	September 29, 2026
Curriculum Revisions	SBTE Diploma Rev 2021 & Rev 2026 (R26)	Auditing System	Carmel-Linx Academic Platform Engine
Applicable Programs	Civil, Mechanical, Electrical, Automobile, Computer	Report Status	FOR ACADEMIC COUNCIL & HOD REVIEW

1. EXECUTIVE OVERVIEW & COMPARATIVE MATRIX

Course Category	Revision 2021 (R21 Scheme - Unified %-to-Mark Logic)	Revision 2026 (R26 Scheme - Standardized)
Theory Courses	10 Marks out of 50 CIA (20% of CIA) Direct % to Mark : 95-100% → 10M, 85-94% → 9M, 75-84% → 8M, ..., 5-14% → 1M, 0-4% → 0M.	5 Marks out of 40 CIE (12.5% of CIE) SBTE Table 2.1 : ≥90% → 5M, 80-89% → 4M, 75-79% → 3M, 70-74% → 2M, 65-69% → 1M, <65% → 0M.
Practical / Lab Courses	15 Marks out of 75 CIA (20% of CIA) Proportional : Actual % converted to 15M (100% → 15.0M, 50% → 7.5M). <i>Faculty manual override supported in PracticalEvaluations.</i>	5 Marks out of 60 CIE (8.3% of CIE) SBTE Table 2.1 : Slabs applied uniformly (5M, 4M, 3M, 2M, 1M, 0M).
Practicum: Basic Science (Physics, Chemistry, etc.)	<i>Not defined in R21 (treated as separate Theory + Lab courses).</i>	5 Marks out of 40 CIE (12.5% of CIE) SBTE Table 2.1 : Shared between 45h lecture + 45h lab hours.
Practicum: Program Core (Integrated Technical Subjects)	<i>Not defined in R21 (treated as separate Theory + Lab courses).</i>	5 Marks out of 40 CIE (12.5% of CIE) SBTE Table 2.1 : Standardized across all engineering branches.
Engineering Drawing / Graphics	10 Marks out of 50 CIA (20% of CIA) Direct % to Mark : 95-100% → 10M, 85-94% → 9M, 75-84% → 8M, ..., 0-4% → 0M.	5 Marks out of 60 CIE (8.3% of CIE) SBTE Table 2.1 : Slabs applied uniformly (5M, 4M, 3M, 2M, 1M, 0M).
Seminar Courses	7.5 Marks out of 75 CIA (10% of CIA) Proportional : Actual % converted to 7.5M (100% → 7.5M, 50% → 4.0M).	<i>Curriculum roadmap incorporates seminars into later R26 semesters under Table 2.1 standard.</i>
Major Project / Capstone	15 Marks out of 75 CIA (20% of CIA) Proportional : Actual % converted to 15M (100% → 15.0M, 50% → 7.5M).	<i>Capstone Project adheres to standardized Table 2.1 CIE mapping.</i>
Health & Physical Education (Audit Course)	<i>Not applicable in R21 curriculum.</i>	5 Marks out of 60 CIE (8.3% of CIE) SBTE Table 2.1 . <i>Note: Excluded from CO Attainment calculations.</i>

2. DETAILED AUDIT: REVISION 2021 (R21) SYSTEM LOGIC (UPDATED)

Key Core Rules for Revision 2021:

- Strictly Based on Actual Conducted & Attended:** Attendance percentage is calculated strictly on actual sessions conducted and actual sessions attended ($(\text{actualAttended} / \text{actualConducted}) \times 100$), never on proposed hours.
- Direct Percentage-to-Marks Conversion:** Raw attendance marks are computed as $(\text{percentage} / 100) \times \text{maxAttendanceMarks}$.
- Rounding Rule:** Decimal values adjust to nearest: **.5 and above → next whole/half number; below .5 → lower number.**
- Exam Eligibility Isolated to Tutor Report:** Exam eligibility rules are applicable exclusively to the **consolidated attendance of all subjects together** in the Tutor Report. All individual subject evaluation and marks registers strictly avoid attendance shortage/eligibility status notes.

2.1 Common 10-Mark Attendance Schedule (Theory & Drawing - Max 10 Marks)

Attendance Percentage Range	Awarded Marks (Max 10)	Mathematical Rounding Evaluation
95.00% to 100.00%	10.0 Marks	Raw: 9.50 to 10.00 → Rounds to 10

Attendance Percentage Range	Awarded Marks (Max 10)	Mathematical Rounding Evaluation
85.00% to 94.99%	9.0 Marks	Raw: 8.50 to 9.49 → Rounds to 9
75.00% to 84.99%	8.0 Marks	Raw: 7.50 to 8.49 → Rounds to 8
65.00% to 74.99%	7.0 Marks	Raw: 6.50 to 7.49 → Rounds to 7
55.00% to 64.99%	6.0 Marks	Raw: 5.50 to 6.49 → Rounds to 6
45.00% to 54.99%	5.0 Marks	Raw: 4.50 to 5.49 → Rounds to 5
35.00% to 44.99%	4.0 Marks	Raw: 3.50 to 4.49 → Rounds to 4
25.00% to 34.99%	3.0 Marks	Raw: 2.50 to 3.49 → Rounds to 3
15.00% to 24.99%	2.0 Marks	Raw: 1.50 to 2.49 → Rounds to 2
5.00% to 14.99%	1.0 Mark	Raw: 0.50 to 1.49 → Rounds to 1
0.00% to 4.99%	0.0 Marks	Raw: 0.00 to 0.49 → Rounds to 0

2.2 Laboratory, Major Project & Seminar Attendance Scaling

Course Category	Max Att. Marks	At 100% Att.	At 50% Att.	Calculation Formula
Practical / Lab Courses	15.0 Marks	15.0 Marks	7.5 Marks	$\text{round}(((\text{Present} / \text{Conducted}) * 15) * 2) / 2$
Major Project (Clause 11.2.5)	15.0 Marks	15.0 Marks	7.5 Marks	$\text{round}(((\text{Present} / \text{Conducted}) * 15) * 2) / 2$
Seminar (Clause 11.2.6)	7.5 Marks	7.5 Marks	4.0 Marks	$\text{round}(((\text{Present} / \text{Conducted}) * 7.5) * 2) / 2$

3. DETAILED AUDIT: REVISION 2026 (R26) STANDARDIZED SCHEME

In the Revision 2026 curriculum, the State Board of Technical Education (SBTE) introduced a completely standardized system. **All courses adhere to Table 2.1 with a maximum of 5 Marks for attendance**, eliminating inter-subject formula discrepancies.

3.1 Standard SBTE Table 2.1 Attendance Mark Schedule

Student Attendance Range	CIE Marks Awarded (Max 5)	SBTE Academic Status	Regulatory Action Required
≥ 90.00%	5 Marks	Eligible	Direct Hall Ticket Generation
80.00% to 89.99%	4 Marks	Eligible	Direct Hall Ticket Generation
75.00% to 79.99%	3 Marks	Eligible	Regular ESE Eligibility (Minimum threshold)
70.00% to 74.99%	2 Marks	Condonation Required	Condonation application via Institutional Principal
65.00% to 69.99%	1 Mark	Condonation Required	Condonation fee & genuine medical/extenuating grounds
50.00% to 64.99%	0 Marks	Special Condonation	Requires SBTE Board approval; else student is detained
< 50.00%	0 Marks	Detained	Repeat semester/course; strictly ineligible for ESE

3.2 Course-wise CIE Integration in Revision 2026

Course Classification	Total CIE	ESE Marks	Att. Marks	Complete CIE Component Breakdown
Theory Courses	40 Marks	60 Marks	5 Marks	Self-Learning Table 2.2 (15M) + Series Exams Avg (20M) + Attendance (5M)
Practicum: Basic Science (Physics, Chemistry)	40 Marks	60 Marks	5 Marks	Continuous Practical (10M) + Theory Series (10M) + Practical Series (10M) + Self-Learning (5M) + Attendance (5M)
Practicum: Program Core (Branch-specific Practicum)	40 Marks	60 Marks	5 Marks	Continuous Practical (10M) + Theory Series (10M) + Practical Series (10M) + Self-Learning (5M) + Attendance (5M)
Practical Courses (Pure Lab subjects)	60 Marks	40 Marks	5 Marks	Continuous Day Work (30M) + Open Ended OEE (10M) + Practical Series (15M) + Attendance (5M)
Engineering Drawing	60 Marks	40 Marks	5 Marks	Slot Evaluations CE (30M) + CA1 & CA2 Tests (15M) + Open Ended (10M) + Attendance (5M)
Health & Physical Ed (Audit Course)	60 Marks	40 Marks	5 Marks	Continuous Activity (30M) + Fitness Tests CA1/CA2 (15M) + Viva/Logbook (10M) + Attendance (5M)

4. TECHNICAL ARCHITECTURE & DATABASE AUDITING

Dual Data-Flow Pipeline: The platform maintains synchronization between real-time teacher hourly logs (class_logs_attendance) and the centralized student academic registry (student_attendance).

Database Entity / Table	Fields Analyzed	Operational Role in Attendance Processing
class_logs_attendance	present_students (JSON), absent_students (JSON), batch_no, period_no	Primary operational log recorded hourly by subject teachers in classrooms and labs. Used directly for Rev 2021 Theory & Practical computations.
student_attendance	reg_no, subject_code, status ('Present', 'Late', 'Absent'), date	Individual student normalized attendance ledger. Used for all Revision 2026 modules and Rev 2021 Drawing/Seminar/Project courses.
practical_evaluations	attendance_marks (decimal nullable)	Stores faculty manual override marks for R21 laboratory courses where specific makeup labs or concessions were officially approved.
r21_drawing_attendance_evaluations	attendance_marks, percentage	Stores finalized attendance marks and approved condonations for R21 Drawing.

5. CRITICAL AUDIT OBSERVATIONS FOR ACADEMIC REVIEW

1. Discrepancy in R21 Between Theory and Other Courses:

In Revision 2021, students in Theory receive **0 Marks** if their attendance falls below 75%. However, in Drawing, Seminar, and Major Project, students between 65% and 75% receive partial attendance marks (20% to 40% of the maximum weight). This was by SBTE R21 design, but created confusion among staff.

2. R21 Lab Attendance Linear vs Slabs:

R21 Practical calculates attendance as a direct linear fraction: $(\text{Present} / \text{Total}) \times 15$. A student with 89% receives 13.4M, whereas in R21 Theory an 89% student falls into the 9M stepdown slab.

3. Complete Unification in Revision 2026 (R26):

Revision 2026 completely resolves prior discrepancies. Every subject type strictly implements **Table 2.1 (Max 5 Marks)**. Outcome attainment calculations also correctly isolate academic competencies by subtracting attendance marks where required.

6. EVALUATION, CORRECTION & ACADEMIC SIGN-OFF BLOCK

This section is intended for physical or digital endorsement by the Academic Committee, Heads of Departments, and Exam Cell.

Internal Auditor / System Administrator Remarks:

All formulas verified against SBTE curriculum regulations and source controllers. Logic verified clean.

Academic Council / Exam Cell Correction Directive:

- [] Verified & Approved without corrections
- [] Corrections requested (specify below)

Verified By System Auditor Carmel-Linx Academic Engine	Academic Council Coordinator Signature & Date	Head of Department (HOD) Department Seal & Date	Principal / Vice Principal Carmel Polytechnic College
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